Ewald points of monotone polytopes: the minimum in dimension eight and branching simplex bundles

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Abstract

A monotone polytope is a smooth reflexive lattice polytope. Its Ewald set $\mathcal{E}(P) = P \cap (-P) \cap \mathbb{Z}^D$ consists of the lattice points of P whose antipodes also lie in P. Crespo, Pelayo and Santos asked for the minimum $E_{\min}(D)$ of $|\mathcal{E}(P)|$ over monotone D-polytopes. They determined it for $D \leq 7$ and gave upper bounds from towers of small simplex bundles, with exponential growth rate $9^{1/3}$; in dimension eight their bound is 531. We prove $E_{\min}(8) = 507$ by counting the Ewald points of all 749,892 monotone 8-polytopes with two independent programs. The minimum is attained by exactly three polytopes: a tower of simplex bundles that branches at two different facets, and two toric blow-ups of it with the same Ewald set. The same description covers all ten monotone 8-polytopes with at most 531 Ewald points. With explicit certificates, the same computation verifies Ewald's conjecture in dimension eight, in the strong form that each facet contains a lattice basis consisting of Ewald points. For towers whose bundles may be attached at arbitrary facets we give a composition formula expressing Ewald counts through two conditional counts per facet. It yields two infinite families that improve the known upper bounds in all dimensions $D \equiv 0, 2 \pmod 3$ with $D \geq 8$, the best towers in every dimension up to 30, and the asymptotic bound $\lim_D E_{\min}(D)^{1/D} \leq 2.0628 < 9^{1/3}$. In the other direction, every such tower over a segment has at least $3 \cdot 2.03^{D-1}$ Ewald points.

1 Introduction

1.1 Monotone polytopes and Ewald's conjecture

Let $P \subset \mathbb{R}^D$ be a full-dimensional lattice polytope with the origin in its interior. It is *reflexive* if it has an inequality description

$$P = \{x \in \mathbb{R}^D : \ \langle u_G, x \rangle \leq 1 \text{ for every facet } G\},$$

with primitive integral outer normals u_G, and *smooth* if at every vertex the normals of the facets through that vertex form a basis of $\mathbb{Z}^D$. Smooth reflexive polytopes are called *monotone*: they are the moment polytopes of monotone symplectic toric manifolds [6], and the polar duals of smooth Fano polytopes, i.e. the anticanonical polytopes of smooth toric Fano varieties [2]. Up to unimodular equivalence there are finitely many in each dimension. They have been classified by Øbro's algorithm [7, 8] up to dimension eight, and by Lorenz and Paffenholz in dimension nine [5]; there are 749,892 of dimension eight and 8,229,721 of dimension nine.

The *Ewald set* of a monotone polytope is

$$\mathcal{E}(P) = P \cap (-P) \cap \mathbb{Z}^D = \{m \in \mathbb{Z}^D : \ |\langle u_G, m \rangle| \leq 1 \text{ for every facet } G\}.$$

Ewald's conjecture [4] asserts that $\mathcal{E}(P)$ contains a basis of $\mathbb{Z}^D$. It has been verified by computer for $D \le 7$ by Øbro [8], who checked the stronger statement that $\mathcal{E}(P) \cap G$ contains a lattice basis for every facet G (see [3, Theorem 3.9]). McDuff [6, p. 134] mentions that Øbro verified this strong form up to dimension eight, but in Øbro's thesis it is stated only up to dimension seven [3, Remark 3.10]. In general it is not even known whether $\mathcal{E}(P) \neq \{0\}$ [9, 6, 3]. This weakest form was checked by computer in dimension at most nine: Øbro (for $D \le 8$) and Paffenholz (for $D = 9$) verified that every facet of every monotone polytope contains a nonzero Ewald point [1, Section 3.4]. It is also known for several structural classes [3, 10, 11].

1.2 The minimum number of Ewald points

Crespo, Pelayo and Santos [3, Question 6.3] asked: *What is the minimum of $|\mathcal{E}(P)|$ in each dimension, and which polytope attains it?* Write

$$E_{\min}(D) = \min\{|\mathcal{E}(P)| : \ P \text{ a monotone } D\text{-polytope}\}.$$

They computed $|\mathcal{E}(P)|$ for all monotone polytopes of dimension at most seven, which gives

$$E_{\min}(1), \dots, E_{\min}(7) = 3,\ 7,\ 13,\ 27,\ 59,\ 117,\ 243, \tag{1}$$

and they remarked that the minimum could well be 1 in some dimension. They also introduced the *small fibre bundle* of a monotone polytope at a facet [3, Definition 6.4] and iterated it at the newly created facet to obtain [3, Corollary 6.7, Table 5]

$$E_{\min}(n) \le \begin{cases} 3 \cdot 9^k, & n = 3k+1, \\ \frac{59}{9} \cdot 9^k, & n = 3k+2,\ k \ge 1, \\ 13 \cdot 9^k, & n = 3k+3, \end{cases} \tag{2}$$

so that $\limsup E_{\min}(D)^{1/D} \le 9^{1/3} \approx 2.0801$; in dimension eight the bound is 531. The definition in [3] allows a bundle at *any* facet of the base, but the iteration in [3, Lemmas 6.5–6.6] always uses the newest one. This paper studies what happens when bundles are attached at arbitrary facets, and determines the minimum in dimension eight.

1.3 Results

A *tower* is a polytope obtained from a segment (or a point) by repeatedly attaching simplex bundles, each at an arbitrary facet of the current polytope; see Definition 2.4. All towers are monotone.

Theorem A (Dimension eight)**.** $E_{\min}(8) = 507$. *Up to unimodular equivalence exactly three monotone 8-polytopes have 507 Ewald points: the tower*

$$T = \left\{ (x,y,z,w) \in \mathbb{R} \times \mathbb{R}^3 \times \mathbb{R}^2 \times \mathbb{R}^2 : \begin{array}{l} |x| \le 1, \quad y_i, z_j, w_j \ge -1, \quad y_1+y_2+y_3+3x \le 1, \\ z_1+z_2-2y_1 \le 1, \quad w_1+w_2-2y_2 \le 1 \end{array} \right\}$$

and two polytopes obtained from T by a toric blow-up along a torus-invariant subvariety of codimension two, which leaves the Ewald set unchanged. More generally, the monotone 8-polytopes with at most 531 Ewald points have $|\mathcal{E}(P)| \in \{507, 521, 531\}$, with 3, 4 and 3 polytopes respectively; each of them is a tower or a single such blow-up of a tower with the same Ewald set.

The tower T is obtained from the segment by a small Δ_3-bundle at one facet, followed by small Δ_2-bundles at two *different* fibre facets of that Δ_3. The corresponding step in [3] attaches both Δ_2-bundles in succession at the newest facet and produces 531 points.

Theorem B (Ewald's conjecture in dimension eight)**.** *For every monotone 8-polytope P and every facet G of P, the set $\mathcal{E}(P) \cap G$ contains a basis of $\mathbb{Z}^8$. In particular, Ewald's conjecture holds in dimension eight.*

The published verification of Øbro covers $D \leq 7$. McDuff's remark quoted above indicates that the case $D = 8$ had also been checked, but we could not find an account of that computation. Theorem B supplies one, with an explicit certificate for every polytope and every facet (Section 6.3).

The constructions behind Theorem A extend to all dimensions.

Theorem C (Two infinite families)**.** *For every $k \geq 2$,*

$$E_{\min}(3k+2) \leq 507 \cdot 9^{k-2} < 531 \cdot 9^{k-2}, \qquad E_{\min}(3k+3) \leq 1033 \cdot 9^{k-2} < 1053 \cdot 9^{k-2}.$$

The right-hand sides are the bounds (2).

Theorem D (Exponential rate)**.** *The limit $\lim_{D\to\infty} E_{\min}(D)^{1/D}$ exists and*

$$\lim_{D\to\infty} E_{\min}(D)^{1/D} \leq \left(19195 + 4\sqrt{2297138}\right)^{1/14} = 2.06279697\ldots < 9^{1/3}.$$

Optimizing over towers gives much better values in every fixed dimension; Table 1 lists the best towers for $D \leq 30$. In every dimension $8 \leq D \leq 30$ except $D = 10$ they improve (2), by 19% at $D = 30$. Finally, towers cannot do better than a fixed exponential rate.

Theorem E (A barrier for towers)**.** *Let $\lambda = 203/100$. Every tower P over the segment of dimension D satisfies $|\mathcal{E}(P)| \geq 3\lambda^{D-1}$, and every tower over a point satisfies $|\mathcal{E}(P)| \geq \lambda^D$. More generally, if P is a tower of dimension D over a monotone d-polytope B_0, then at least λ^{D-d} points of $\mathcal{E}(P)$ project to the origin of B_0.*

Hence the growth rate of towers lies in $[2.03, 2.0628]$; the best towers we found grow by a factor of about 2.055 per dimension (Remark 4.2).

D	best tower	bound (2)	ratio	D	best tower	bound (2)	ratio
1	3	3	1.000	16	167 955	177 147	0.948
2	7	7	1.000	17	345 247	387 099	0.892
3	13	13	1.000	18	710 377	767 637	0.925
4	27	27	1.000	19	1 469 505	1 594 323	0.922
5	59	59	1.000	20	3 006 733	3 483 891	0.863
6	117	117	1.000	21	6 121 753	6 908 733	0.886
7	243	243	1.000	22	12 746 715	14 348 907	0.888
8	507	531	0.955	23	26 001 367	31 355 019	0.829
9	1 033	1 053	0.981	24	53 122 417	62 178 597	0.854
10	2 187	2 187	1.000	25	110 129 209	129 140 163	0.853
11	4 447	4 779	0.931	26	224 474 623	282 195 171	0.795
12	9 187	9 477	0.969	27	456 642 313	559 607 373	0.816
13	19 065	19 683	0.969	28	969 651 721	1 162 261 467	0.834
14	39 253	43 011	0.913	29	1 973 060 839	2 539 756 539	0.777
15	80 593	85 293	0.945	30	4 086 342 827	5 036 466 357	0.811

Table 1: Ewald counts of the best towers (Section 3.3) compared with the bounds (2); for $D = 2$, which (2) does not cover, the bound column shows $E_{\min}(2) = 7$. For $D \leq 7$ both columns equal the exact minima (1), and for $D = 8$ the tower value is the exact minimum (Theorem A).

Methods. The basic observation is that every Ewald point has all facet heights in $\{-1, 0, 1\}$, and that a bundle attached at a facet sees the rest of the polytope only through the height of that facet. Section 2 turns this into a composition formula: each facet carries a pair (u, v) of conditional counts, attachments at the same facet multiply, and a simplex bundle combines the pairs of its new facets through a Laurent polynomial. Theorems C and D follow from explicit branching patterns (Sections 3 and 4). Theorem E uses a piecewise linear potential that is supermultiplicative under both operations (Section 5). Theorem A is a complete computation over the classification, carried out by two independent programs (Section 6).

2 Simplex bundles, towers and conditional counts

2.1 Simplex bundles

Throughout, $\Delta_n = \{y \in \mathbb{R}^n : \ y_j \geq -1, \ y_1 + \cdots + y_n \leq 1\}$ is the monotone n-simplex; $\Delta_1 = [-1, 1]$.

Definition 2.1. Let $B \subset \mathbb{R}^d$ be monotone, let G be a facet of B with primitive outer normal ℓ, and let $n \geq 1$ and $0 \leq q \leq n$ be integers. Put

$$S_{n,q}(B, G) = \Big\{(x, y) \in \mathbb{R}^d \times \mathbb{R}^n : \ x \in B, \ \ y_j \geq -1 \ (1 \leq j \leq n), \ \ \sum_{j=1}^{n} y_j + q\langle \ell, x\rangle \leq 1\Big\}.$$

For $q = n$ this is the small fibre bundle with base B and fibre Δ_n at G of [3, Definition 6.4]; for $q = 0$ it is the product $B \times \Delta_n$.

Lemma 2.2. *$P = S_{n,q}(B, G)$ is a monotone polytope of dimension $d + n$. Its facets are the lifts $\{\langle u_H, x\rangle = 1\} \cap P$ of the facets H of B, the n* fibre facets *$\{y_j = -1\}$, and the* sum facet *$\{\sum_j y_j + q\langle \ell, x\rangle = 1\}$, with outer normals $(u_H, 0)$, $(0, -e_j)$ and $(q\ell, 1, \ldots, 1)$.*

Proof. With $z_j = y_j + 1$ the fibre over $x \in B$ is $\{z \geq 0, \ \sum_j z_j \leq n + 1 - q\langle \ell, x\rangle\}$. Since $\langle \ell, x\rangle \leq 1$ on B and $q \leq n$, the right-hand side is at least 1, so every fibre is a full-dimensional simplex whose size is an affine function of x. Hence P is combinatorially $B \times \Delta_n$, all listed inequalities define facets, and the vertices of P are the points (v, y) with v a vertex of B and y a vertex of the fibre over v: either $y = (-1, \ldots, -1)$, or y has one coordinate $n - q\langle \ell, v\rangle$ and all others -1. These are lattice points. At such a vertex the matrix of active normals is block triangular. Its base block is the matrix of normals of B at v, which is unimodular, and its fibre block consists of n of the $n + 1$ vectors $-e_1, \ldots, -e_n, (1, \ldots, 1)$, which has determinant ± 1. Thus P is smooth. All right-hand sides equal 1 and the new normals are primitive, so P is reflexive. □

Remark 2.3. For $q = n$ the $n + 1$ new facets are interchangeable. The linear map of $\mathbb{Z}^d \times \mathbb{Z}^n$ that fixes every $(w, 0)$ and every $(0, -e_j)$ with $j \geq 2$, and sends $(0, -e_1)$ to $(n\ell, 1, \ldots, 1)$, is an involution that exchanges $(0, -e_1)$ and $(n\ell, 1, \ldots, 1)$. Its transpose is therefore a lattice automorphism of $S_{n,n}(B, G)$ that preserves the projection to B, fixes the lifted facets and exchanges the sum facet with the first fibre facet. Together with the permutations of the coordinates y_j these automorphisms realize every permutation of the $n + 1$ new facets. Hence, for small bundles, it does not matter which of the new facets receives a later attachment.

Definition 2.4. Let B_0 be a monotone polytope (the *seed*). A *tower over B_0* is a polytope obtained from B_0 by finitely many operations $B \mapsto S_{n,q}(B, G)$, where at each step G is an arbitrary facet of the current polytope and $n \geq 1$, $0 \leq q \leq n$ are arbitrary. A tower over a point starts with a simplex Δ_n; the segment Δ_1 is the most important seed.

By Lemma 2.2 every tower is monotone, and facets created at any stage remain facets later. A facet may therefore be used again after other bundles have been attached elsewhere, and several bundles may be attached at the same facet. The towers of [3] use $q = n$ and always the newest sum facet. Products of towers are towers: after building one factor, an operation $S_{n,0}$ creates an independent factor Δ_n, and a second tower that starts with Δ_n can be built on it and its descendants. Since $\mathcal{E}(P \times Q) = \mathcal{E}(P) \times \mathcal{E}(Q)$ and products of monotone polytopes are monotone, $E_{\min}(d+e) \le E_{\min}(d)\, E_{\min}(e)$; by Fekete's lemma $\lim_D E_{\min}(D)^{1/D}$ exists and equals $\inf_D E_{\min}(D)^{1/D}$.

2.2 Heights

For a facet G of a monotone polytope P and $m \in \mathcal{E}(P)$ the *height* $h_G(m) = \langle u_G, m\rangle$ lies in $\{-1, 0, 1\}$. Put

$$a_G(P) = \#\{m \in \mathcal{E}(P) : h_G(m) = 1\}, \qquad b_G(P) = \#\{m \in \mathcal{E}(P) : h_G(m) = 0\}.$$

Since $\mathcal{E}(P) = -\mathcal{E}(P)$, also $a_G(P) = \#\{h_G = -1\}$ and $|\mathcal{E}(P)| = 2a_G(P) + b_G(P)$. We call G *balanced* if $a_G(P) = b_G(P)$. This is the condition $|\mathcal{E}_+(P,G)| = |\mathcal{E}_0(P,G)|$ of [3, Lemma 6.6].

Lemma 2.5 (Conservation)**.** *Let $P = S_{n,q}(B, G)$. A point (x, y) lies in $\mathcal{E}(P)$ if and only if $x \in \mathcal{E}(B)$ and the new heights*

$$h_j = -y_j \ (1 \le j \le n), \qquad h_0 = \sum_j y_j + q\, h_G(x)$$

lie in $\{-1, 0, 1\}$. In that case $h_0 + h_1 + \cdots + h_n = q\, h_G(x)$. Conversely, every $x \in \mathcal{E}(B)$ and every $(h_0, \dots, h_n) \in \{-1, 0, 1\}^{n+1}$ with $\sum_i h_i = q\, h_G(x)$ give exactly one point of $\mathcal{E}(P)$, namely $y_j = -h_j$.

Proof. By Lemma 2.2, $(x, y) \in \mathcal{E}(P)$ means that all heights of (x, y), old and new, lie in $[-1, 1]$. The old heights are those of x, and the new ones are as displayed. The rest is immediate. □

2.3 The facet forest and the composition formula

In a tower, every operation is attached at one facet, its *parent*, and creates $n + 1$ new facets, its *children*: the sum facet and the n fibre facets. For a facet N of the final polytope let $\mathrm{Br}(N)$ be the set of operations attached at N, together with all operations attached (recursively) at facets created by them, and let $\mathrm{Var}(N)$ be the coordinates introduced by these operations. By Lemma 2.5 and induction, the Ewald conditions involving facets created in $\mathrm{Br}(N)$ depend on the other coordinates only through h_N. For $t \in \{-1, 0, 1\}$ let $c_N(t)$ be the number of assignments of $\mathrm{Var}(N)$ that satisfy these conditions when $h_N = t$. Negating all coordinates shows $c_N(1) = c_N(-1)$; we write

$$u_N = c_N(\pm 1), \qquad v_N = c_N(0),$$

and call (u_N, v_N) the *conditional counts* of N. Similarly $c_O(t)$ denotes the count for a single operation O together with everything attached below its children.

Proposition 2.6 (Composition)**.** (i) *If nothing is attached at N, then $(u_N, v_N) = (1, 1)$.*

(ii) *If $O_1, \dots, O_r$ are the operations attached directly at N, then $c_N(t) = \prod_i c_{O_i}(t)$; in particular $u_N = \prod_i u_{O_i}$ and $v_N = \prod_i v_{O_i}$.*

(iii) *If $O = S_{n,q}(\cdot, N)$ has children N_0 (sum facet) and $N_1, \dots, N_n$, then*

$$c_O(t) = [z^{qt}] \prod_{i=0}^{n} (v_{N_i} + u_{N_i}(z + z^{-1}));$$

that is, $u_O = [z^q]\Phi_O$ and $v_O = [z^0]\Phi_O$ with $\Phi_O = \prod_i (v_{N_i} + u_{N_i}(z + z^{-1}))$.

(iv) *If P is a tower over B_0, then*

$$|\mathcal{E}(P)| = \sum_{m\in\mathcal{E}(B_0)} \prod_{G \text{ facet of } B_0} c_G(h_G(m)).$$

In particular, if B_0 is the segment and (u, v) is the product of the conditional counts of its two facets, then $|\mathcal{E}(P)| = 2u+v$. If B_0 is a point and O is the initial simplex, then $|\mathcal{E}(P)| = c_O(0)$.

Proof. (i) is clear. In (ii) the variable sets of different operations attached at N are disjoint, and each operation interacts with the others only through h_N. For (iii), Lemma 2.5 says that, for fixed $h_N = t$, the new coordinates of O correspond to the vectors $(h_0, \dots, h_n) \in \{-1, 0, 1\}^{n+1}$ with $\sum_i h_i = qt$, and the variables below N_i then contribute $c_{N_i}(h_i)$ independently. Summing $\prod_i c_{N_i}(h_i)$ over these vectors is the stated coefficient extraction, because $v + u(z + z^{-1})$ records weight v for height 0 and u for heights ±1. For $t = -1$ the symmetry of Φ_O gives the same value as for $t = 1$. (iv) follows in the same way from the description of $\mathcal{E}(P)$ in Lemma 2.5. The two facets $\pm x \le 1$ of the segment have heights $\pm x$; since c_G is even, their contributions combine to (u, v), and the three Ewald points $x = -1, 0, 1$ of the segment contribute u, v, u. For a point seed, the initial simplex is attached to a "facet" of height 0. □

A bare simplex attachment has conditional counts

$$u = [z^q](z^{-1} + 1 + z)^{n+1}, \qquad v = [z^0](z^{-1} + 1 + z)^{n+1}; \tag{3}$$

for $q = n$ these are $(2, 3)$, $(3, 7)$, $(4, 19)$, $(5, 51)$ for $n = 1, 2, 3, 4$. We write $Q(z) = z^{-1} + 1 + z$ and $L_n(z) = v + u(z + z^{-1})$ for the leaf with $q = n$, so $L_2 = 7 + 3(z + z^{-1})$ and $L_3 = 19 + 4(z + z^{-1})$.

Corollary 2.7. *Let B be monotone and G a facet of B. If a collection of operations with conditional counts (u, v) is attached at G (and nothing else is attached), the result P satisfies $|\mathcal{E}(P)| = 2a_G(B)\, u + b_G(B)\, v$.*

Proof. Apply Proposition 2.6(iv) with seed B; only G carries attachments. □

Lemma 2.8 (Unimodality)**.** *For real $0 \le r_i \le 1$ the coefficients of $\prod_i(1+r_i(z+z^{-1}))$ are symmetric and nonincreasing away from z^0. Consequently every facet of a tower has $0 < u_N \le v_N$.*

Proof. Let (c_k) be symmetric and nonincreasing for $k \ge 0$, and multiply by $1 + r(z + z^{-1})$. The new coefficients satisfy $d_k - d_{k+1} = r(c_{k-1} - c_{k+2}) + (1 - r)(c_k - c_{k+1}) \ge 0$ for $k \ge 0$, using $c_{-1} = c_1 \ge c_2$ when $k = 0$. For the second statement, (i) and (ii) of Proposition 2.6 preserve $0 < u \le v$. In (iii), divide each factor by v_{N_i} and apply the first statement with $r_i = u_{N_i}/v_{N_i}$; then $[z^q] \le [z^0]$, and $[z^q] > 0$ because $q \le n$ and all $u_{N_i} > 0$. □

Remark 2.9. For $q = n$ and a single operation at a facet G, the slices of the new sum facet are $a' = n\, a + |\mathcal{E}_+(\Delta_n)|\, b$ and $b' = 2a + |\mathcal{E}_0(\Delta_n)|\, b$, where (a, b) are the slices of G and $(|\mathcal{E}_+(\Delta_n)|, |\mathcal{E}_0(\Delta_n)|)$ are the slices of any facet of Δ_n. This is [3, Lemma 6.5]. For $n = 3$ one gets $(a', b') = (3a+6b,\ 2a+7b)$. Hence a balanced facet stays balanced, with $a' = b' = 9a$, and $|\mathcal{E}|$ is multiplied by 9 [3, Lemma 6.6].

3 Branching constructions

3.1 The polytope T

The tower T of Theorem A is the segment with a $S_{3,3}$ attached at its facet $x \le 1$, and with $S_{2,2}$ attached at the fibre facets $-y_1 \le 1$ and $-y_2 \le 1$ of that Δ_3. By Proposition 2.6, the facet $x \le 1$

has conditional counts given by

$$Q(z)^2 L_2(z)^2 = 387 + 320(z + z^{-1}) + 178(z^2 + z^{-2}) + 60(z^3 + z^{-3}) + 9(z^4 + z^{-4}),$$

namely $(u, v) = (60, 387)$. The other facet of the segment has $(1, 1)$, so

$$|\mathcal{E}(T)| = 2 \cdot 60 + 387 = 507.$$

For comparison, attaching the second Δ_2 at the sum facet of the first one, as in [3, Lemma 6.6(iii)], gives 531.

3.2 Two branching operations at a balanced facet

Proposition 3.1. *Let B be a monotone d-polytope with a balanced facet G.*

(i) *Attach $S_{3,3}$ at G, and $S_{2,2}$ at two distinct fibre facets of this Δ_3. The result P has dimension $d + 7$ and $|\mathcal{E}(P)| = 169\,|\mathcal{E}(B)|$.*

(ii) *Attach $S_{2,2}$ at G, and $S_{3,3}$ at both fibre facets of this Δ_2. The result P has dimension $d + 8$ and $|\mathcal{E}(P)| = \frac{1033}{3}|\mathcal{E}(B)|$.*

Proof. Let $a = a_G(B) = b_G(B)$, so $|\mathcal{E}(B)| = 3a$. In (i) the attached collection has $\Phi = Q^2 L_2^2$, so $(u, v) = (60, 387)$ as above. In (ii),

$$Q(z) L_3(z)^2 = 697 + 561(z + z^{-1}) + 168(z^2 + z^{-2}) + 16(z^3 + z^{-3}),$$

so $(u, v) = (168, 697)$. By Corollary 2.7, $|\mathcal{E}(P)| = a(2u + v)$, which is $507a = 169 \cdot 3a$ in (i) and $1033a$ in (ii). □

Proof of Theorem C. Let T_{3j+1} be the segment followed by j small Δ_3-bundles, each at the sum facet of the previous one (the first at $x \le 1$). Its last sum facet is balanced with $a = b = 9^j$, by Remark 2.9 and induction from the segment ($a = b = 1$), so $|\mathcal{E}(T_{3j+1})| = 3 \cdot 9^j$. Applying Proposition 3.1 to T_{3j+1} gives monotone polytopes of dimensions $3j + 8$ and $3j + 9$ with $169 \cdot 3 \cdot 9^j = 507 \cdot 9^j$ and $1033 \cdot 9^j$ Ewald points. Put $k = j + 2$. □

Remark 3.2. In the iteration of [3] the corresponding seven- and eight-dimensional steps multiply $|\mathcal{E}(B)|$ by $9 \cdot \frac{59}{3} = 177$ and by $\frac{13}{3} \cdot 81 = 351$. Branching replaces these by 169 and $\frac{1033}{3} \approx 344.3$. The gain comes from the joint distribution of the fibre heights. Over a balanced facet, the heights (h_1, h_2) of two fibre facets of a Δ_3-bundle are uniformly distributed on $\{-1, 0, 1\}^2$, so two further bundles at these facets act independently.

3.3 Best towers up to dimension thirty

Proposition 2.6 reduces the Ewald count of a tower over a point to a finite recursion on pairs (u, v), and all its operations are monotone in their inputs. One can therefore minimize $|\mathcal{E}|$ over all towers of a given dimension exactly, by dynamic programming over Pareto-minimal sets. Concretely, we compute for each m the Pareto-minimal pairs (v, u) of collections that use m coordinates and hang at one facet. Products of such collections give the attachments at one facet, and formula (iii) combines them into simplex operations. We did this for all towers over a point whose fibres have dimension at most 4, with arbitrary attachment facets and twists $0 \le q \le n$. Allowing fibre dimension up to 6 gives the same values for all $D \le 28$. The resulting minima are the column "best

tower" of Table 1, and the minimizing trees are listed in Appendix A. All of them use $q = n$ only, i.e. small bundles in the sense of [3]. The supplementary files continue the computation to $D = 36$.

Every value in Table 1 is an upper bound for $E_{\min}(D)$, witnessed by an explicit tower. We checked these values independently of Proposition 2.6. For $8 \le D \le 19$ we built the witness polytopes explicitly and counted their Ewald points by direct enumeration. For $D \le 10$, an exhaustive search over all towers with arbitrary fibre dimensions gives the same minima; in dimension 8 this includes towers whose twists are arbitrary lattice points $p \in nB^\vee$ rather than multiples of one facet normal. We do not claim optimality of the table beyond the class described.

4 An exponential rate below $9^{1/3}$

Lemma 4.1 (Bridge)**.** *Let G be a facet of a monotone polytope B, with slices (a, b). Attach $S_{2,2}$ at G, attach a collection with conditional counts (u, v) at its sum facet, and let G' be the fibre facet $-y_1 \le 1$. Then*

$$\begin{pmatrix} a_{G'} \\ b_{G'} \end{pmatrix} = K(u,v) \begin{pmatrix} a \\ b \end{pmatrix}, \qquad K(u,v) = \begin{pmatrix} u+v & u+v \\ 2u & 2u+v \end{pmatrix}.$$

Proof. Fix a point of $\mathcal{E}(B)$ with $h_G = t$, and count completions with $h_1 = s$. By Lemma 2.5 the remaining heights satisfy $h_0 + h_2 = 2t - s$, with weight v or u for $h_0 = 0$ or ± 1 and weight 1 for h_2. For $s = 1$ this gives $u + v$ if $t \in \{1, 0\}$ and 0 if $t = -1$; for $s = 0$ it gives u if $t = \pm 1$ and $v + 2u$ if $t = 0$. Summing over the a, b, a points with $t = 1, 0, -1$ gives the formula. □

Proof of Theorem D. At the sum facet of the bridge attach $S_{3,3}$, and at each of its three fibre facets attach a further $S_{3,3}$. This collection uses 12 coordinates and has $\Phi = Q(z)L_3(z)^3$, and

$$Q(z)L_3(z)^3 = 17731 + 14119(z + z^{-1}) + 5500(z^2 + z^{-2}) + 976(z^3 + z^{-3}) + 64(z^4 + z^{-4}),$$

so $(u, v) = (976, 17731)$. Together with the two coordinates of the bridge we obtain an operation that adds 14 dimensions and has transfer matrix

$$K = K(976, 17731) = \begin{pmatrix} 18707 & 18707 \\ 1952 & 19683 \end{pmatrix}.$$

The output facet G' is a facet of the new polytope, so the operation can be repeated. Starting from the segment and its facet $x \le 1$ (slices $(1, 1)$), k repetitions give monotone polytopes P_k of dimension $14k + 1$ with

$$|\mathcal{E}(P_k)| = e_k = (2, 1)\, K^k \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad e_0 = 3, \quad e_1 = 96463, \quad e_{k+2} = 38390\, e_{k+1} - 331693817\, e_k.$$

The eigenvalues of K are $\rho, \sigma = 19195 \pm 4\sqrt{2297138}$ with $0 < \sigma < \rho$, and $e_k = A\rho^k + (3 - A)\sigma^k$ with $A = (96463 - 3\sigma)/(\rho - \sigma) > 0$. Hence $e_k^{1/(14k+1)} \to \rho^{1/14}$. Products with segments give $E_{\min}(14k + 1 + r) \le 3^r e_k$ for $0 \le r \le 13$, so $\limsup E_{\min}(D)^{1/D} \le \rho^{1/14}$; the limit exists by Fekete's lemma. Finally $16 \cdot 2297138 = 36754208 < 6063^2$ gives $\rho < 25258$, and $25258^3 = 16113759473512 < 22876792454961 = 9^{14}$, so $\rho^{1/14} < 9^{1/3}$. □

Remark 4.2. (a) The simplest gadget of this kind uses a bare Δ_3 at the sum facet of the bridge: $K(4, 19) = \left(\begin{smallmatrix} 23 & 23 \\ 8 & 27 \end{smallmatrix}\right)$ adds five dimensions and already gives the rate $(25+2\sqrt{47})^{1/5} = 2.0776\ldots < 9^{1/3}$. (b) The family P_k beats (2) only eventually: $e_1 = 96463$ exceeds the bound 85293 in dimension 15, while $e_4 = 1864620809452020987 < 1951230258860988573$ in dimension 57. The best towers of

Table 1 are much smaller. For instance 80593 in dimension 15 and 3006733 in dimension 20, where (2) gives 3483891. (c) Among towers that repeat a single gadget with one marked output facet, all fibres of dimension at most 4, the 14-dimensional operation above has the smallest rate for gadgets of dimension at most 16; the next best rates are 2.0646 (dimension 16) and 2.0659 (dimension 13). We found this by the same dynamic programming applied to "one-hole" collections. Nevertheless the best towers grow more slowly. Let t_D be the smallest Ewald count of a tower of dimension D with fibres of dimension at most 4 (Table 1 for $D \le 30$; the computation extends to $D \le 36$). For $28 \le D \le 36$ the ratios $(t_D/t_{D-14})^{1/14}$ lie between 2.0528 and 2.0596, which suggests that branching into several repeated subtrees beats every single repeated gadget. We do not have a proof.

5 A lower bound for towers

For $0 \le u \le v$ with $v > 0$ define

$$F(u,v) = \min\Big\{\frac{96v+399u}{400},\ \frac{91v+209u}{300}\Big\} = v\,f(u/v), \qquad f(r) = \min\Big\{\frac{6}{25}+\frac{399}{400}r,\ \frac{91}{300}+\frac{209}{300}r\Big\}.$$

The two lines meet at $r = \tau = 4/19$, and

$$f(0) = \tfrac{6}{25}, \quad f(\tau) = \tfrac{9}{20}, \quad f(1) = 1, \qquad \tfrac{6}{25} \le f(r) \le 1, \qquad 3f(r) \le 1 + 2r \quad (0 \le r \le 1), \quad (4)$$

as one checks at the breakpoints $0, \tau, 1$. F is homogeneous of degree one and nondecreasing in both arguments.

Lemma 5.1 (Parallel attachments)**.** *For $0 \le u_i \le v_i$ with $v_i > 0$,*

$$F(u_1u_2, v_1v_2) \ge F(u_1,v_1)\,F(u_2,v_2).$$

Proof. By homogeneity it suffices to show $f(rs) \ge f(r)f(s)$ for $r, s \in [0,1]$, i.e. $L(rs) \ge f(r)f(s)$ for each of the two lines L. On each rectangle cut out by $\{0, \tau, 1\}$ both $f(r)$ and $f(s)$ are affine, so $L(rs) - f(r)f(s)$ is bilinear there and attains its minimum at a corner. At the corners $(r,s) \in \{0,\tau,1\}^2$ the values of $(L_1(rs) - f(r)f(s),\ L_2(rs) - f(r)f(s))$ are $(0,0) \mapsto (\frac{114}{625}, \frac{1843}{7500})$, $(0,\tau) \mapsto (\frac{33}{250}, \frac{293}{1500})$, $(0,1) \mapsto (0, \frac{19}{300})$, $(\tau,\tau) \mapsto (\frac{621}{7600}, \frac{1001}{7600})$, $(\tau,1) \mapsto (0,0)$, $(1,1) \mapsto (\frac{19}{80}, 0)$, and symmetric; all are nonnegative. □

Lemma 5.2 (Simplex expansion)**.** *Let $\lambda = 203/100$. For $n \ge 1$, $0 \le q \le n$ and $0 \le u_i \le v_i$, $v_i > 0$ $(0 \le i \le n)$, let $u = [z^q]\Phi$ and $v = [z^0]\Phi$ with $\Phi = \prod_i (v_i + u_i(z + z^{-1}))$. Then*

$$F(u,v) \ge \lambda^n \prod_{i=0}^{n} F(u_i, v_i).$$

Proof. By homogeneity take $v_i = 1$ and $r_i = u_i \in [0,1]$. By Lemma 2.8, $[z^q]\Phi \ge [z^n]\Phi$, and F is nondecreasing, so it suffices to treat $q = n$.

Large n. The central coefficient v is the largest of the $2n+3$ coefficients of Φ, whose sum is $\prod_i (1 + 2r_i) \ge \prod_i 3f(r_i)$ by (4). Hence

$$F(u,v) \ge \tfrac{6}{25}v \ge \frac{6 \cdot 3^{n+1}}{25(2n+3)} \prod_i f(r_i).$$

For $n = 9$ the factor exceeds λ^9, since $6 \cdot 3^{10} \cdot 100^9 - 25 \cdot 21 \cdot 203^9 > 0$. When n increases by one the factor grows by $3(2n+3)/(2n+5) \ge \lambda$, because $300(2n+3) - 203(2n+5) = 194n - 115 > 0$. This proves the claim for all $n \ge 9$.

Small n. For $1 \le n \le 8$ and either defining line L_j of F, the difference $v\,L_j(u/v) - \lambda^n \prod_i f(r_i)$ is affine in each r_i separately on each box cut out by $\{0, \tau, 1\}$. Indeed u and v are multi-affine in the r_i, and so is $\prod_i f(r_i)$ on each box. Its minimum is therefore attained at a corner. By symmetry a corner is described by the numbers (α, β, γ) of coordinates equal to $0, \tau, 1$, with $\alpha + \beta + \gamma = n + 1$. At such a corner $u = U/19^\beta$ and $v = V/19^\beta$, where

$$U = [z^n](4z^{-1} + 19 + 4z)^\beta Q(z)^\gamma, \qquad V = [z^0](4z^{-1} + 19 + 4z)^\beta Q(z)^\gamma,$$

and $\prod_i f(r_i) = (6/25)^\alpha (9/20)^\beta$. Table 2 lists, for each n, the minimum over all 216 corners of the two ratios

$$C_1 = \frac{(96V + 399U)/400}{19^\beta (6/25)^\alpha (9/20)^\beta}, \qquad C_2 = \frac{(91V + 209U)/300}{19^\beta (6/25)^\alpha (9/20)^\beta}.$$

Every entry is at least λ^n, which is checked by exact rational arithmetic. □

n	$\min C_1$	at (α, β, γ)	$\min C_2$	at (α, β, γ)	λ^n
1	543/200	$(0, 0, 2)$	691/300	$(0, 0, 2)$	2.030
2	25/6	$(2, 0, 1)$	316/75	$(0, 0, 3)$	4.121
3	4648000/555579	$(0, 3, 1)$	171/20	$(0, 0, 4)$	8.365
4	6891/400	$(0, 0, 5)$	2843/150	$(0, 0, 5)$	16.982
5	1593/40	$(0, 0, 6)$	939/20	$(0, 0, 6)$	34.473
6	40521/400	$(0, 0, 7)$	18613/150	$(0, 0, 7)$	69.980
7	13683/50	$(0, 0, 8)$	102409/300	$(0, 0, 8)$	142.060
8	60987/80	$(0, 0, 9)$	28753/30	$(0, 0, 9)$	288.382

Table 2: The finite certificate for Lemma 5.2. The tightest case is $n = 3$, where $\min C_1 = 4648000/555579 = 8.3660\ldots$ exceeds $\lambda^3 = 8.3654\ldots$ by a factor 1.00007.

Proof of Theorem E. By induction over the facet forest, using Proposition 2.6, Lemma 5.1 and Lemma 5.2, every facet N whose branch uses m coordinates satisfies $F(u_N, v_N) \ge \lambda^m$; a facet with nothing attached has $F(1, 1) = 1$. For the segment seed, the two facets combine by Lemma 5.1 to a pair (u, v) with $m = D - 1$, and by (4) $|\mathcal{E}(P)| = 2u + v \ge 3F(u, v) \ge 3\lambda^{D-1}$. For a point seed, the initial simplex O satisfies $F(u_O, v_O) \ge \lambda^D$ by Lemma 5.2, and $|\mathcal{E}(P)| = v_O \ge F(u_O, v_O)$ because $f \le 1$. For a general seed B_0, the points over the origin of B_0 have all seed heights 0, so their number is $\prod_G v_G \ge \prod_G F(u_G, v_G) \ge \lambda^{D-d}$. □

Remark 5.3. Theorem E concerns towers only: bundles with fibres other than simplices, twists by several facet normals, or seeds of unbounded dimension are not covered. The potential is nearly tight: by the case $n = 3$ of Table 2, Lemma 5.2 with this F fails for $\lambda > (4648000/555579)^{1/3} = 2.03005\ldots$, so a substantial improvement of 2.03 requires a different potential.

6 Dimension eight

6.1 Data and method

We used the classification of monotone 8-polytopes produced by Øbro's algorithm [7], in the facet-list format distributed by Lorenz and Paffenholz [5] (file `smooth_fano_8d`). Each of the 749,892 polytopes is given by its facet inequalities $1 + \langle a, x \rangle \ge 0$, so that $u = -a$. The SHA-256 checksum of the decompressed file we used is `ec78d1b6f0f1207310c9e4c6d693ba5f6802de75fa14ec5d46aeca81ccd808ab`.

The count rests on a simple observation, already used in the proof of [3, Proposition 6.1]. If $u_1, \dots, u_D$ are the normals of the facets through a vertex of a monotone polytope P, they form a basis B of $\mathbb{Z}^D$, and every $m \in \mathcal{E}(P)$ has $Bm \in \{-1,0,1\}^D$. Hence $\mathcal{E}(P) = \{B^{-1}t : \ t \in \{-1,0,1\}^D,\ |\langle u_G, B^{-1}t\rangle| \le 1\ \forall G\}$, and $|\mathcal{E}(P)|$ is computed exactly by testing 3^D candidates. We used two independent implementations.

(1) A vectorized Python program. It first checks that each polytope in the file has, for every i, the normal e_i or $-e_i$; this holds for all 749,892 polytopes. It then tests the 6561 points of $\{-1,0,1\}^8$ in integer-valued arithmetic.

(2) A C program with its own parser. It searches for 8 facet normals with determinant ± 1, transforms by the inverse matrix (checked exactly in integers), and enumerates t depth-first with pruning.

Both programs agree on all 749,892 counts. As a check, the maximum is $6561 = 3^8$, attained only by the cube, in accordance with [3, Proposition 6.1]. Both programs reproduce $|\mathcal{E}(T)| = 507$ and the count 531 of the corresponding polytope from [3].

6.2 Results

The smallest values of $|\mathcal{E}(P)|$ and their multiplicities are

$\lvert\mathcal{E}(P)\rvert$	507	521	531	549	553	559	567	575	579	581	587	591	595	597	601
number	3	4	3	3	3	3	7	1	1	3	1	3	1	1	1

Proof of Theorem A. By the computation, $E_{\min}(8) = 507$, attained by three polytopes of the list, which have 12, 13 and 13 facets. The tower T is monotone (Lemma 2.2), has 12 facets and 507 Ewald points, so it is equivalent to the unique minimizer with 12 facets.

Let P' be one of the two minimizers with 13 facets. In each case the computation exhibits a facet whose normal is $u_1 + u_2$, where u_1, u_2 are the normals of two facets G_1, G_2 of P'. Deleting this facet gives a polytope P which we checked, by enumerating all vertices, to be monotone with 12 facets, adjacent facets G_1, G_2, and the same 507 Ewald points. By uniqueness $P \cong T$. In a monotone polytope, a vertex $w \notin G$ has $\langle u_G, w\rangle \le 0$, because the value is an integer below 1. So the vertices of P with $\langle u_1 + u_2, w\rangle > 1$ are exactly those of the codimension-two face $G_1 \cap G_2$, and $P' = P \cap \{\langle u_1 + u_2, x\rangle \le 1\}$ is obtained by truncating this face. Dually, the normal fan of P' is the stellar subdivision of the fan of P along the cone spanned by u_1, u_2, so the toric variety of P' is the blow-up of that of P along the torus-invariant subvariety corresponding to $G_1 \cap G_2$. The Ewald set is unchanged because $|\langle u_1 + u_2, m\rangle| \le 1$ already holds on $\mathcal{E}(P)$; the counts agree.

The same argument applies to the values 521 and 531. In each case exactly one polytope has 12 facets. It is equivalent to the tower obtained from the segment as follows: for 521, a Δ_2 at one facet, a Δ_2 at one of its fibre facets, and a Δ_3 at its sum facet; for 531, the tower of [3]. The remaining polytopes have 13 facets and arise from these by the same kind of truncation, with unchanged Ewald set. □

The three minimizers, in the coordinates of the data file, are listed in Appendix B.

6.3 Ewald's conjecture in dimension eight

Proof of Theorem B. For each of the 749,892 polytopes we enumerated $\mathcal{E}(P)$ as above. We then searched for a lattice basis among $\mathcal{E}(P)$ itself and among $\mathcal{E}(P) \cap G$ for each facet G (10,878,172

facets in total). The search is greedy: vectors are added one at a time as long as the chosen set stays primitive, i.e. extendable to a basis of $\mathbb{Z}^8$. Primitivity is tested by maintaining a unimodular matrix U with (chosen vectors) $U = [L \mid 0]$ and L lower triangular with diagonal ± 1; a candidate p is accepted if and only if the entries of pU outside the first k columns have greatest common divisor 1, and U is then updated by integer column operations. The candidates are taken in random order, with restarts. In every case a basis was found. Each of the $749{,}892 + 10{,}878{,}172 = 11{,}628{,}064$ bases was then certified independently of the search, by computing its determinant exactly (fraction-free Gaussian elimination in 64-bit integers); it is ± 1 in every case. □

The certificate is constructive. A failure of the greedy search would not have been a counterexample, but no failure occurred.

7 Remarks and questions

(1) *Dimension nine.* The same programs apply verbatim to the 8,229,721 monotone 9-polytopes of [5], with 3^9 candidates each. The best tower gives $E_{\min}(9) \leq 1033$, below the bound 1053 of (2).

(2) *Are towers optimal?* For $D \leq 8$ the minimum is attained by a tower, and in dimension eight every polytope with at most 531 Ewald points is a tower or a blow-up of one with the same Ewald set. Is $E_{\min}(D)$ attained by a tower in every dimension?

(3) *The growth rate of towers* lies in $[2.03, 2.0628]$, and the best towers we found suggest a value near 2.055 (Remark 4.2). Determine it.

(4) *Lower bounds.* No nontrivial lower bound for $E_{\min}(D)$ is known in general. Even $E_{\min}(D) \geq 3$, i.e. $\mathcal{E}(P) \neq \{0\}$ for all P, is open for $D \geq 10$ [3, 11]; for $D \leq 9$ it follows from the computations reported in [1, Section 3.4]. Theorem E gives exponential lower bounds only for towers.

A Optimal trees

A tower over a point is written as a tree. D_n denotes a small Δ_n-bundle ($q = n$). The entries in brackets are the bundles attached at its facets: "s:" at its sum facet, and each "f:" at a different fibre facet. The root D_1 is the segment, whose two facets are interchangeable. By Remark 2.3 only the number of new facets carrying attachments matters, not which ones; for instance the tree for $D = 8$ is the tower T of Theorem A.

D	$\lvert\mathcal{E}\rvert$	tree
1	3	D_1
2	7	D_2
3	13	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_2]$
4	27	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_3]$
5	59	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_2[\mathrm{s}{:}\,\mathrm{D}_2]]$
6	117	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_2]]$
7	243	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_3]]$
8	507	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_2,\ \mathrm{f}{:}\,\mathrm{D}_2]]$
9	1 033	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_2[\mathrm{s}{:}\,\mathrm{D}_3,\ \mathrm{f}{:}\,\mathrm{D}_3]]$
10	2 187	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_3]]]$
11	4 447	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_2[\mathrm{s}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_2],\ \mathrm{f}{:}\,\mathrm{D}_3]]$
12	9 187	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_2[\mathrm{s}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_3],\ \mathrm{f}{:}\,\mathrm{D}_3]]$
13	19 065	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_2[\mathrm{s}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_2,\ \mathrm{f}{:}\,\mathrm{D}_2],\ \mathrm{f}{:}\,\mathrm{D}_3]]$
14	39 253	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_2[\mathrm{s}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_3,\ \mathrm{f}{:}\,\mathrm{D}_2],\ \mathrm{f}{:}\,\mathrm{D}_3]]$
15	80 593	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_2[\mathrm{s}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_3,\ \mathrm{f}{:}\,\mathrm{D}_3],\ \mathrm{f}{:}\,\mathrm{D}_3]]$
16	167 955	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_2[\mathrm{s}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_3,\ \mathrm{f}{:}\,\mathrm{D}_2,\ \mathrm{f}{:}\,\mathrm{D}_2],\ \mathrm{f}{:}\,\mathrm{D}_3]]$
17	345 247	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_2[\mathrm{s}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_3,\ \mathrm{f}{:}\,\mathrm{D}_2,\ \mathrm{f}{:}\,\mathrm{D}_3],\ \mathrm{f}{:}\,\mathrm{D}_3]]$
18	710 377	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_2[\mathrm{s}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_3,\ \mathrm{f}{:}\,\mathrm{D}_3,\ \mathrm{f}{:}\,\mathrm{D}_3],\ \mathrm{f}{:}\,\mathrm{D}_3]]$
19	1 469 505	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_2[\mathrm{s}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_3,\ \mathrm{f}{:}\,\mathrm{D}_3],\ \mathrm{f}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_2,\ \mathrm{f}{:}\,\mathrm{D}_2]]]$
20	3 006 733	$\mathrm{D}_1[\mathrm{s}{:}\,\mathrm{D}_2[\mathrm{s}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_3,\ \mathrm{f}{:}\,\mathrm{D}_3],\ \mathrm{f}{:}\,\mathrm{D}_3[\mathrm{s}{:}\,\mathrm{D}_3,\ \mathrm{f}{:}\,\mathrm{D}_2]]]$

Trees for $21 \le D \le 30$ are included in the supplementary files.

B The minimizers in dimension eight

Each polytope is listed by the vectors a of its facet inequalities $1 + \langle a, x\rangle \ge 0$, as in [5]; the index is the position in the file (starting at 0).

Index 39293 (the tower T, 12 facets).

$(0,0,0,0,0,-1,0,0)$ $(0,0,0,0,-1,0,0,0)$ $(0,-1,0,0,0,0,0,0)$ $(-2,0,0,1,0,0,1,0)$
$(0,0,0,0,0,0,0,1)$ $(-1,0,0,0,0,0,0,0)$ $(0,0,0,-1,0,0,0,0)$ $(0,0,0,0,0,0,-1,0)$
$(0,0,0,0,0,0,0,-1)$ $(0,-2,1,0,0,1,0,0)$ $(0,0,-1,0,0,0,0,0)$ $(1,1,0,0,1,0,0,3)$

Index 38568 (13 facets).

$(-2,0,0,1,0,0,0,1)$ $(0,0,0,0,0,-1,0,0)$ $(0,-2,1,0,0,0,1,0)$ $(0,0,-1,0,0,0,0,0)$
$(0,0,0,-1,0,0,0,0)$ $(0,-1,0,0,1,0,0,0)$ $(0,0,0,0,0,0,0,-1)$ $(0,-1,0,0,0,0,0,0)$
$(-1,0,0,0,0,0,0,0)$ $(0,0,0,0,0,0,-1,0)$ $(1,1,0,0,-3,1,0,0)$ $(0,0,0,0,-1,0,0,0)$
$(0,0,0,0,1,0,0,0)$

Index 38660 (13 facets).

$(0,0,0,0,0,-1,0,0)$ $(1,1,1,0,0,0,-3,0)$ $(0,-2,0,1,0,0,0,1)$ $(-1,0,0,0,0,0,0,0)$
$(0,0,-1,0,0,0,1,0)$ $(-2,0,0,0,1,1,0,0)$ $(0,0,0,-1,0,0,0,0)$ $(0,-1,0,0,0,0,0,0)$
$(0,0,-1,0,0,0,0,0)$ $(0,0,0,0,-1,0,0,0)$ $(0,0,0,0,0,0,-1,0)$ $(0,0,0,0,0,0,0,-1)$
$(0,0,0,0,0,0,1,0)$

C Supplementary files

The supplement contains the programs used here and the outputs of all runs: the two Ewald counters (Python and C) with the per-polytope counts in dimension eight, the certified basis search

for Theorem B, the structure checks for Theorem A, the dynamic programming over towers with the optimal trees for $D \leq 30$, the exhaustive tower searches, the brute-force checks of the witness polytopes, and the exact verification of Section 5. The data file itself is not included; its source and checksum are given in Section 6.

The supplement consists of the 43 files of the archive `ewald_supplement.tar.xz` (1,671,796 bytes) with SHA-256

`957ac35a396ce4b8d117666d34458684d957bd3e91fc2674e3bdb90d787713c2`

Appendix D prints this archive as two text files, each a JavaScript bookmarklet that carries half of it in base64. Running both bookmarklets in the same browser tab, in either order, checks the SHA-256 and saves the archive. Without a browser, concatenate the strings `D="..."` of parts 1 and 2 and decode the result from base64.

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D Supplementary machine-readable material

This appendix contains the two bookmarklet files described in Appendix C, parts 1 and 2 of 2, minified. Each is printed between the lines ==BEGIN:*name*== and ==END:*name*==, in lines of at most 256 characters. No line begins or ends with a space. Joining the lines of each file exactly, for example with

```
pdftotext ewald_points.pdf - | tr -d '\f' |
  awk '/^==BEGIN:/{f=substr($0,9,length($0)-10);next} /^==END:/{f="";next} f{printf "%s",$0 > f}'
```

The same blocks appear verbatim in the LaTeX source of this paper. Some text extractors drop the spaces of such very small type; the base64 strings don't. "..." contain no spaces, so they survive. The SHA-256 checksums at the end confirm it.

```
==BEGIN:ewald_supplement_bookmarklet_part1of2.txt==
javascript:/*ewald_supplement_bookmarklet_part1of2.txt*/(function(){var P=1,N=2,F="ewald_supplement.tar.xz",H="957ac35a396c4e4b81d117666d34586864d957bd3e91fc2674e3bdb90d787713c2",D="/Td6WFoAAATm1rRGAgAhAHwAAAAQc1jM5kEW77/9dADKdyEZa2i+DVpOmcCwQHuHbHrR2rvWZITOKNpquYvOCGiW8+NAB+AEEN/IozOJNnSBHZPCIpEskbhGLDwE+3TmAPmMzyUIv2cpbjpnaoCYm0W8JYZKlmA3PEuK+wvoiyQo03IJS94xsFdu5ew7P2V26NLTx4D/sPpYlJyOUvjFN3JzQY37tvvEEdV0ixxa+pTz6GvG7UAOc/uiiRmTZrvXj3S/mmxy3IufRQ/AR88o899jqUc2+Zkia7ieZXvTCd1806S1eDb72gcPvhWQSS37kRYOxdqSnjzl/pzdG8AHjv4sADmV7EzhSf4uIfvPv4oqecxTnLVDMUGIYzcJ1MIKEMEGDiP0CrQEIxiNiCf3/RAOOSY4geTrIeKpnOHdjzVBvd2IDFxAnEN+aaOtwvCpEJgVCOg/o++AVQ+a4HDMV7fah5f4uIfvPLyoqecxTnLVDMUGIYzcJ1MIKEMEGDiP0CrQEIxiNiCf3/RAOOSY4geTrIeKpnOHdjzVBvd2IDFxAnEN+aaOtwvCpEJgVCOg/o++AVQ+a4HDMV7fah5f4uIfvPLyoqecxTnLVDMUGIYzcJ1MIKEMEGDiP0CrQEIxiNiCf3/RAOOSY4geTrIeKpnOHdjzVBvd2IDFxAnEN+aaOtwvCpEJgVCOg/o++AVQ+/s4hLDv7dB+Zq8nBZ1Ud4lJ8W5IuF9DkrrpJ1J/RnYRCeAbE+TmVqyyUJAdgrxoGBykSAgEl71v7/FI1hKztMyQ5DRMGNJC7SFZkLaoU+DzWxQRbFa+6EM5SwGoigiuqKoIxfWrHgXOimyWSJcOLuIGvWgaEP2qJ2gHtCIAYGwp8XNxRwVRCVFd1fdlOYduZeKE+MZDNgIyTS+PoJ9jrnZ2CkTcECVBE8Sw1ZqIvSdvwapuKpzKzeRN7fvInqyyJU
```

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